

Stuart Davis

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EDUCATION

University of Melbourne

2022 – Jun 2027

M. MechEng. (Aerospace Specialisation), WAM 82 / H1

B. Sci. (Mechanical Engineering Major), WAM 83 / H1

- Relevant coursework: Fluid Dynamics (94), Thermodynamics (91), Systems Modelling & Analysis (89), Aerospace Propulsion (current), Aerospace Dynamics & Control (82)

PROJECTS

Computationally Designed Liquid Bipropellant Engine

Dec 2025 – Jun 2026

Winner, Best LOx Bipropellant — Race2Space 2026

Capstone, 3-person team

- Designed, built, and hot fired an additively manufactured (DMLM CuCrZr) regen engine: 35 bar chamber, 40-channel helical counter-flow cooling, 12-element bi-centrifugal swirl injector. Ground-up in 7 months.
- Developed [BRUV](#), a C/C#/Python program sizing, simulating, optimising, and then generating all print-ready geometry directly from targets with embedded AM constraints — *no manual CAD for either printed part*.
- Achieved 5.4 kN peak thrust across five hot fires (≈ 31 s total), including $\pm 25\%$ throttle and OFR sweeps. Steady thrust, pressure, and exhaust plume; bi-swirl internal-mixing injector showed no ablation, soot, or tarnishing post-teardown.
- Manufactured in partnership with CSIRO Lab22. Live interviewed on ABC News.

Hybrid Rocket Simulator & Optimiser

2025

- Hybrid rocket simulator ([BRO](#)) written in C with a Python GUI front-end, modelling tank blowdown, injector flow, fuel regression, combustion, and trajectory through to apogee; NASA CEA and CoolProp precompiled to static lookup tables to keep the inner loop fast under optimisation.
- Optimisation search with Powell's method over a nine-term cost function; derivative-free by necessity, since five terms are hard manufacturability and safety penalties that leave the surface non-smooth.
- Sole program used to design and size Genesis, ARES' 10,000 ft-class hybrid motor for Project Odyssey.

Propulsion Ground Support Equipment

2024 – present

- Co-led design and fabrication of a 2.5 kN mobile static-fire stand for hybrid engine testing, sized by FEA and hand calculations to a 5 safety factor for the 500 N subscale motor; assembled and hot fired successfully.
- Operator of the team's mobile DAQ and control cart across the test campaign: 7 thermocouple, 12 pressure transducer, and 4 load cell channels, driving 7 valves and pyrotechnic firing output.

EXPERIENCE

ARES — University of Melbourne Rocketry Club

2024 – Present

Propulsion Lead

Melbourne, Australia

- Lead a propulsion subteam of 19 across engine design, tank and fluid systems, ground support equipment, and thrust structure — covering project direction, design review, test readiness, and safety sign-off.
- Currently leading Genesis to first hot fire, with V2 development already underway for July 2027: a dual-phase injector characterisation campaign and in-house development of a thermal protection system, coaxial MOV, solenoid vent valve, and quick disconnect.

Safety Dave

Jun 2026 – Present

Firmware Engineer (Part-time)

Melbourne, Australia

- Sole developer of STM32 firmware for the Safe-T-Mover, a production caravan mover. Shipping within 3 weeks with 57 pre-orders.
- Rebuilt the firmware from scratch after inheriting an unfinished codebase, delivering a larger feature set in a smaller binary — small enough to reflash existing boards that could not fit the original build.
- Designed and tuned the control stack: an asymmetric current limiter (proportional clamp-down, constant-rate recovery, under 10% overshoot), PI RPM matching between motors, and stall detection on obstruction or overload.

SKILLS

Programming: C, C++, Python, C#, MATLAB, Java, Git
CAD: SolidWorks, Onshape, Technical drawings, GD&T

Analysis: SolidWorks FEA, Custom solvers, Thermo/fluids
Test Engineering: Custom DAQ/control, STM32, LabJack, Synnax